

Ayush Urs

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EDUCATION

University of California, San Diego

San Diego, CA

M.S. Computer Science

University of California, San Diego

San Diego, CA

B.S. Mathematics – Computer Science

GPA: 3.89 September 2023 – March 2026

Relevant Coursework: Machine Learning: Algorithms, Data Structures, Stochastic Processes, Recommender Systems

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, Dart, MATLAB, Stata

Frameworks: Apache Spark, Databricks, Spark SQL, TensorFlow, PyTorch, Keras, LangChain, FastAPI, Flutter

Tools: PostgreSQL, MongoDB, SQLite, Docker, Git, Google Cloud Platform, Vertex AI, Nautilus, VizTracer, HPC, TPU/V100 GPU, Jupyter, VS Code, Cursor, LoRA, MCP(Model Context Protocol)

EXPERIENCE

AI Researcher

Remote

University of Wisconsin–Madison

June 2026 – Present

- Designing multi-agent **LangChain** pipeline routing agents for literature parsing, code generation, and debugging to translate papers into parameterized **PhysiCell/Morpheus** simulations, grounding sweeps in **NCBI** literature.
- Extending a **RAG** retrieval layer over simulation-library Java API docs and reference manuals to improve code-generation accuracy, and scoping an **MCP server** to standardize tool access across the agent pipeline.

Machine Learning Researcher

San Diego, CA

Halcioğlu Data Science Institute

March 2026 – June 2026

- Analyzed **SkyDiscover**, an **LLM-driven** algorithm-discovery framework, and designed a novel **bandit-based selection** strategy with **divide-and-conquer decomposition** to more efficiently allocate compute toward promising solution candidates, outperforming its existing **EoH** and **FunSearch**-style approaches.
- Evaluated the strategy across **200+** optimization tasks, orchestrating **LLM** program generation over a **SQLite** strategy database, developing **unit tests** and using **VizTracer** for real-time execution profiling.

Machine Learning Researcher

San Diego, CA

Human-centered eXtended Intelligence (UCSD Lab)

September 2025 – Present

- Built Reddit-based data pipelines to train LLMs toward emotionally regulated responses, collecting **100K+** samples across **CERQ** (cognitive emotion regulation) dimensions using **Vertex AI** for preference-pair generation.
- Fine-tuned **transformer models** using **Direct Preference Optimization(DPO)** on **4x A100** Nautilus clusters, improving sensitivity to human emotional cues across regulated and dysregulated response variants.

Machine Learning Intern

New York, NY (Hybrid)

Myna Voice Labs (Research Collaboration with Princeton PhD Scholars)

August 2025 – April 2026

- Developing AI models for Parkinson's disease detection from speech using **SVMs, CNNs, and Audio Spectrogram Transformers (AST)**, evaluating model robustness under audio perturbations and noise using **Bayesian intervals**, achieving **89% accuracy** with strong cross-linguistic generalization.
- Optimizing audio feature extraction (**MFCC, OpenL3, Whisper**) and exploring **LoRA-based continual learning** and **knowledge distillation** to mitigate catastrophic forgetting in the **AST (~900M parameters)**.

PROJECTS

Skin Lesion Detection System | *PyTorch, YOLOv8, Docker, Kubernetes*

- Led **9-member** team building an end-to-end skin lesion classification pipeline on **HAM10000**, benchmarking a custom **CNN** against **EfficientNet-B0, ResNet18, and YOLOv8s**, reaching **81.75%** top-1 accuracy.
- Built **GPU training infrastructure** with **Docker** and **Kubernetes** on **A100** nodes, supporting multi-GPU training, persistent checkpoint storage, and interactive debugging pods.

RAG Pipeline for Internal Query Answering | *LangChain, sentence-transformers, MongoDB, FastAPI*

- Built **RAG** pipeline to replace manual internal doc search, using **LangChain** and **sentence-transformers**.

- Designed data layer with **MongoDB Vector Search** for embeddings and document storage, orchestrated via **FastAPI** and deployed with **Docker**, sustaining **100+** concurrent requests at **p95 latency of 1–2 seconds**.

AI-Integrated Island for Eutrophication Control | *Reinforcement Learning, Python*

- Developed AI-driven organic island research prototype using Bellman equation to autonomously target high-phosphate regions, achieving **75% phosphate reduction** in lab trials.
- Implemented value-state matrix for intelligent navigation across grid-based lake environments.